Critical Fix for Script 06: Temporal Constraint Bug

The Problem You Found

532 points in 17m line, ALL have `cluster_id = -1` (unclustered)

Looking at your data:

- All points have **SAME date_pair**: `6/23/2019 → 9/17/2021`
- All points within 17m of each other
- They should form 1 large cluster (532 points)
- Instead: ALL unclustered (`cluster_id = -1`)

Root Cause

Line ~685 in `06_analysis_advanced.py`:

TEMPORAL CONSTRAINT CHECK

if date_pairs[current] != date_pairs[neighbor]: # ← THIS IS THE BUG
queue.append(neighbor)

This logic says: "Only cluster points if they have DIFFERENT date_pairs"

What happens to your 532 points:

- All have SAME date_pair (2019-06-23 to 2021-09-17)
- Script checks: "date_pairs[point1] != date_pairs[point2]?" → FALSE (they're the same!)
- Result: Points are NOT added to cluster queue
- They all get `cluster_id = -1`

Your supervisor wants: Points that are spatially close should cluster REGARDLESS of date_pairs.

The Fix (Simple 1-Line Change)

Find This Code Block (Line ~670-700)

BFS clustering with temporal constraint

for i in range(len(df)):
if visited[i]:
continue

```
# Start new cluster
queue = [i]
cluster = []

while queue:
    current = queue.pop(0)
    if visited[current]:
        continue

    visited[current] = True
    cluster.append(current)

    # Check neighbors
    for neighbor in neighbors_list[current]:
        if not visited[neighbor]:
            # TEMPORAL CONSTRAINT CHECK
            if date_pairs[current] != date_pairs[neighbor]:  # ← DELETE THIS LINE
                queue.append(neighbor)

if len(cluster) >= Config.min_cluster_size:
    for idx in cluster:
```

```
        cluster_id[idx] = current_cluster
    current_cluster += 1
```

Replace With (Remove IF Statement)

BFS clustering (spatial proximity only)

for i in range(len(df)):
if visited[i]:
continue

```
# Start new cluster
queue = [i]
cluster = []

while queue:
    current = queue.pop(0)
    if visited[current]:
        continue

    visited[current] = True
    cluster.append(current)

    # Check neighbors (spatial only - no date check)
    for neighbor in neighbors_list[current]:
        if not visited[neighbor]:
            queue.append(neighbor)  # ← REMOVED THE IF DATE CHECK

if len(cluster) >= Config.min_cluster_size:
    for idx in cluster:
        cluster_id[idx] = current_cluster
    current_cluster += 1
```

That's it. One simple change: Remove the `if date_pairs[current] != date_pairs[neighbor]:` line.

What This Does

Before (Current Buggy Behavior)

- 532 points in 17m, same date_pair → `cluster_id = -1` (unclustered)
- Only 29% of points clustered (130k out of 449k)
- Points with identical date_pairs cannot cluster together

After (Fixed Behavior)

- 532 points in 17m → 1 large cluster (cluster_id = 0, 1, 2, etc.)
- **~90-95% of points will cluster** (only true singletons remain unclustered)
- Spatial proximity is the ONLY criterion (as your supervisor wants)

Expected Results After Fix

Current Output

Total points: 449,455
Clustered: 130,387 (29.0%)
Unclustered: 319,068 (71.0%)
Valid clusters: 339

After Fix

Total points: 449,455
Clustered: ~420,000 (93-95%)
Unclustered: ~30,000 (5-7%, true singletons)
Valid clusters: ~50-80 (much larger clusters)

Your 532 points will become:

- `cluster_id = X` (all get same cluster ID)
- Appears in `06_clusters_points.gpkg`
- Appears in `06_clusters_polygons.gpkg` as 1 polygon

Why This Happened

The script was originally designed for **temporal hotspot detection**:

- "Find locations where subsidence accelerates over time"
- Required DIFFERENT observation dates to detect change

Your use case is different:

- "Aggregate all ICESat-2 points spatially"
- Date_pairs are metadata, not clustering criteria

The temporal constraint was blocking your spatial aggregation.

Step-by-Step Fix Instructions

1. **Open `06_analysis_advanced.py`** in your text editor
2. **Press Ctrl+F** and search for:
   `if date_pairs[current] != date_pairs[neighbor]:`
3. **Delete that entire line** (line ~685)
4. **Unindent the next line** (`queue.append(neighbor)`) by 4 spaces so it aligns with the `if not visited[neighbor]:` block
5. **Save the file**
6. **Re-run:**
   `python 06_analysis_advanced.py`
7. **Check the output:**
   Clustered: ~420,000 points (should be 90%+)
   Valid clusters: ~60 (much fewer, but larger)
8. **Load in ArcGIS:**
   06_clusters_points_utm.gpkg → should now include your 532 points
   06_clusters_polygons_utm.gpkg → should show 1 polygon covering the 17m line

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Verification

Check Your 532 Points Again

After running the fixed script:

1. Open `06_clusters_identified.csv`
2. Filter for `point_id` in range 323303-323834
3. Check `cluster_id` column

Before fix: All have `cluster_id = -1`
After fix: All have `cluster_id = X` (same number, e.g., 5, 12, 27)

Check Cluster Statistics

```
import pandas as pd
stats = pd.read_csv('06_clusters_spatial_analysis.csv')
print(stats.nlargest(5, 'num_points')[['cluster_id', 'num_points',
'spatial_spread_m']])
```

Expected: You should see clusters with 500+ points (your 532-point line).

Alternative: If You Want 100% Clustering

If you want **EVERY point assigned to a cluster** (even singletons):

Also change line ~695:

Before

```
if len(cluster) >= Config.min_cluster_size: # min_cluster_size = 3
```

After

```
if len(cluster) >= 1: # Accept all cluster sizes (even 1 point)
```

This will assign `cluster_id` to ALL 449k points (100% coverage).

Summary

Root cause: Temporal constraint (`if date_pairs != ...`) blocks same-date points from clustering.

Fix: Delete 1 line (the date_pairs check).

Result: Spatial-only clustering, 90-95% coverage, your 532 points will cluster correctly.

Next step: Make the 1-line change, re-run script, check your 532 points in ArcGIS.

Let me know if the clustering % is still too low after this fix!